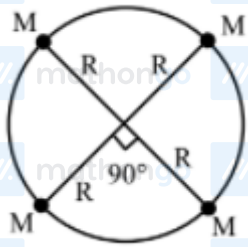


Q1 2021 (01 Sep Shift 2)

Two satellites revolve around a planet in coplanar circular orbits in anticlockwise direction. Their period of revolutions are 1 hour and 8 hours respectively. The radius of the orbit of nearer satellite is 2×10^3 km. The angular speed of the farther satellite as observed from the nearer satellite at the instant when both the satellites are closest is $\frac{\pi}{x} \text{ rad h}^{-1}$ where x is

Q2 2021 (01 Sep Shift 2)

Four particles each of mass M, move along a circle of radius R under the action of their mutual gravitational attraction as shown in figure. The speed of each particle is :



- (1) $\frac{1}{2} \sqrt{\frac{GM}{R(2\sqrt{2}+1)}}$
- (2) $\frac{1}{2} \sqrt{\frac{GM}{R} (2\sqrt{2} + 1)}$
- (3) $\frac{1}{2} \sqrt{\frac{GM}{R} (2\sqrt{2} - 1)}$
- (4) $\sqrt{\frac{GM}{R}}$

Q3 2021 (31 Aug Shift 2)

If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth 'r' below and a height 'r' above the earth surface is :
(Given : $r < R_E$)

- (1) $1 - \frac{r}{R_E} - \frac{r^2}{R_E^2} - \frac{r^3}{R_E^3}$
- (2) $1 + \frac{r}{R_E} + \frac{r^2}{R_E^2} + \frac{r^3}{R_E^3}$
- (3) $1 + \frac{r}{R_E} - \frac{r^2}{R_E^2} + \frac{r^3}{R_E^3}$
- (4) $1 + \frac{r}{R_E} - \frac{r^2}{R_E^2} - \frac{r^3}{R_E^3}$

Q4 2021 (31 Aug Shift 1)

The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses:

$$(1) V = \frac{1}{2} \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

$$(2) V = \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

$$(3) V = \frac{1}{2} \sqrt{\frac{2G(M_1 + M_2)}{r}}$$

$$(4) V = \frac{\sqrt{2G(M_1 + M_2)}}{r}$$

Q5 2021 (27 Aug Shift 2)

A mass of 50 kg is placed at the centre of a uniform spherical shell of mass 100 kg and radius 50 m. If the gravitational potential at a point, 25 m from the centre is $V \text{ kg/m}$. The value of V is :

$$(1) -60 \text{ } \mathrm{G}$$

$$(2) +2G$$

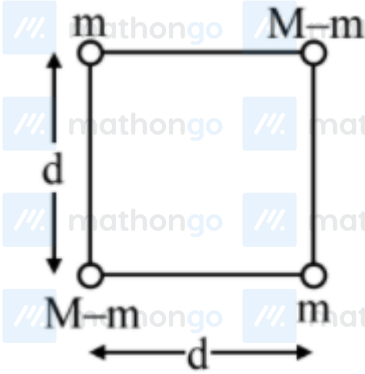
$$(3) -20G$$

$$(4) -4G$$

Q6 2021 (27 Aug Shift 1)

A body of mass $(2M)$ splits into four masses $\{m, M - m, m, M - m\}$, which are rearranged to form a square, as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes

maximum is $x : 1$. The value of x is



Q7 2021 (26 Aug Shift 1)

Inside a uniform spherical shell :

- (a) the gravitational field is zero
- (b) the gravitational potential is zero
- (c) the gravitational field is same everywhere
- (d) the gravitation potential is same everywhere
- (e) all of the above

Choose the most appropriate answer from the options given below :

- (1) (a), (c) and (d) only
- (2) (e) only
- (3) (a), (b) and (c) only
- (4) (b), (c) and (d) only

Answer Key

Q1 (3)

Q2 (2)

Q3 (4)

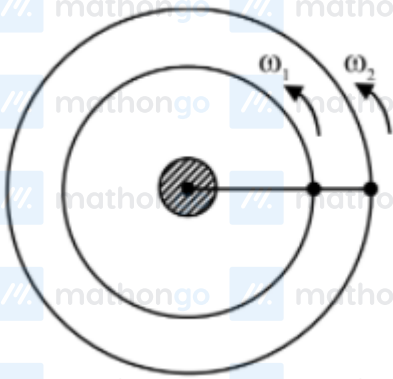
Q4 (2)

Q5 (4)

Q6 (2)

Q7 (1)

Q1 (3)



$$T_1 = 1 \text{ hour}$$

$$\Rightarrow \omega_1 = 2\pi \text{ rad/hour}$$

$$T_2 = 8 \text{ hours}$$

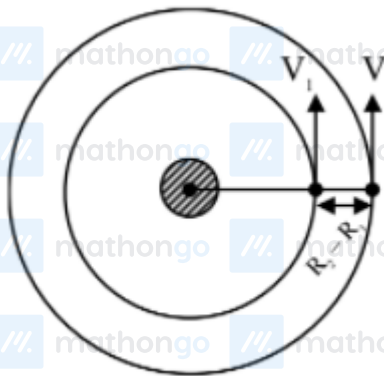
$$\Rightarrow \omega_2 = \frac{\pi}{4} \text{ rad/hour}$$

$$R_1 = 2 \times 10^3 \text{ km}$$

$$\text{As } T^2 \propto R^3$$

$$\Rightarrow \left(\frac{R_2}{R_1}\right)^3 = \left(\frac{T_2}{T_1}\right)^2$$

$$\Rightarrow \frac{R_2}{R_1} = \left(\frac{8}{1}\right)^{2/3} = 4 \Rightarrow R_2 = 8 \times 10^3 \text{ km}$$



$$V_1 = \omega_1 R_1 = 4\pi \times 10^3 \text{ km/h}$$

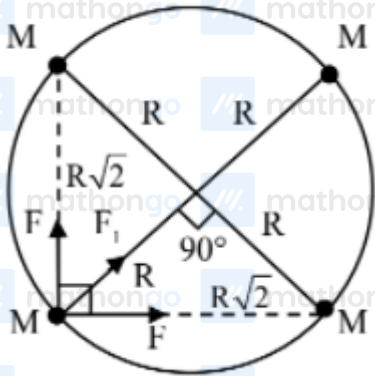
$$V_2 = \omega_2 R_2 = 2\pi \times 10^3 \text{ km/h}$$

$$\text{Relative } \omega = \frac{V_1 - V_2}{R_2 - R_1} = \frac{2\pi \times 10^3}{6 \times 10^3}$$

$$= \frac{\pi}{3} \text{ rad/hour}$$

$$x = 3$$

Q2 (2)



$$F_{\text{net}} = \frac{MV^2}{R}$$

$$\sqrt{2}F + F_1 = \frac{MV^2}{R}$$

$$\sqrt{2} \frac{GMM}{(\sqrt{2}R)^2} + \frac{GMM}{(2R)^2} = \frac{MV^2}{R}$$

$$\frac{GM}{R} \left(\frac{1}{\sqrt{2}} + \frac{1}{4} \right) = V^2$$

$$\frac{GM}{R} \left(\frac{4+\sqrt{2}}{4\sqrt{2}} \right) = V^2$$

$$V = \frac{1}{2} \sqrt{\frac{GM(2\sqrt{2}+1)}{R}}$$

Option (2)

Q3 (4)

$$g_{\text{up}} = \frac{g}{\left(1 + \frac{r}{R}\right)^2}$$

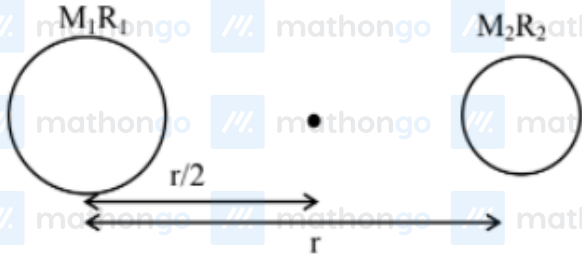
$$g_{\text{down}} = g \left(1 - \frac{r}{R}\right)$$

$$\frac{g_{\text{down}}}{g_{\text{up}}} = \left(1 - \frac{r}{R}\right) \left(1 + \frac{r}{R}\right)^2$$

$$= \left(1 - \frac{r}{R}\right) \left(1 + \frac{2r}{R} + \frac{r^2}{R^2}\right)$$

$$= 1 + \frac{r}{R} - \frac{r^2}{R^2} - \frac{r^3}{R^3}$$

Q4 (2)



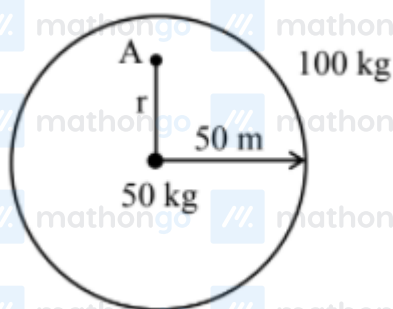
$$\frac{1}{2}mV^2 - \frac{GM_1m}{r/2} - \frac{GM_2m}{r/2} = 0$$

$$\frac{1}{2}mV^2 = \frac{2Gm}{r}(M_1 + M_2)$$

$$V = \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

Option (2)

Q5 (4)



$$V_A = \left[-\frac{GM_1}{r} - \frac{GM_2}{R} \right]$$

$$= \left[-\frac{50}{25}G - \frac{100}{50}G \right]$$

$$= -4G$$

Q6 (2)

Energy is maximum when mass is split equally so

$$\frac{M}{m} = 2$$

Q7 (1)

Inside a spherical shell, gravitational field is zero and hence potential remains same everywhere Hence option

(1)